

Study Designs

**DESCRIPTIVE STUDY DESIGN
AND CASE DEFINITIONS**

Learning Objectives

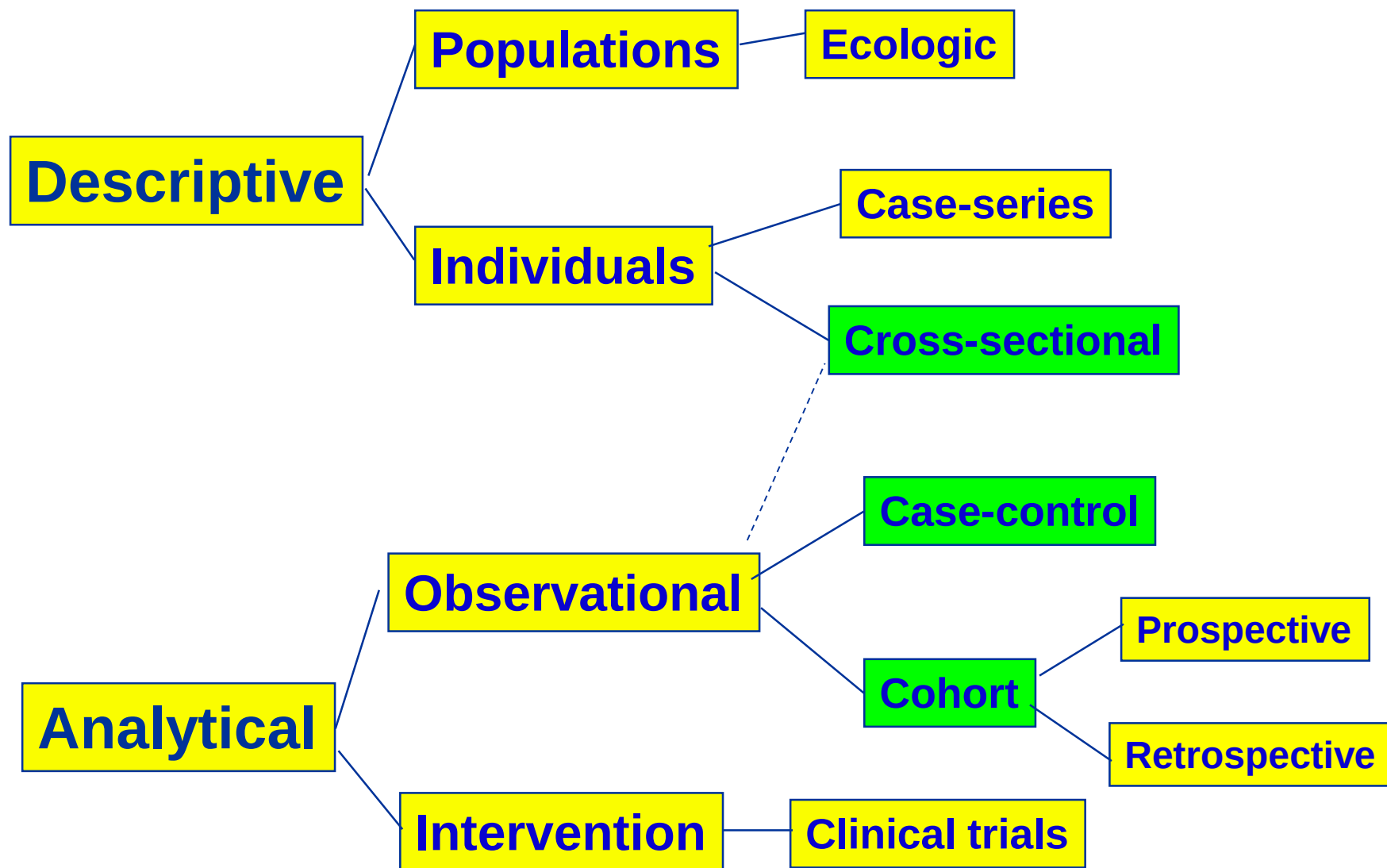
When you have completed this session you will be able to:

- ◆ Understand the difference between analytical and descriptive epidemiology
- ◆ Describe types of descriptive studies

Learning Objectives

- ◆ Explain uses, strengths, weaknesses of descriptive study designs
- ◆ Develop a case definition for surveillance and outbreak investigations

Epidemiological Studies



Descriptive vs Analytic Epidemiology

♦ **Descriptive epidemiology:**

- **generates idea(s) or hypothesis for associations between risk factor and illness**

♦ **Analytical epidemiology:**

- **uses a comparison group to establish an association between risk factors and illness in the two groups**

Descriptive study

“A study concerned with and designed only to describe the existing distribution of variables, without regard to causal or other hypotheses”

Descriptive Studies

- ◆ Used to describe the distribution of disease by time, place, person
- ◆ Useful for hypothesis generation
- ◆ The most frequent design strategy found in the epidemiologic literature

Example

◆ **Some studies simply describe disease/health states/behaviours**

- prevalence of smoking
- rates of lung cancer

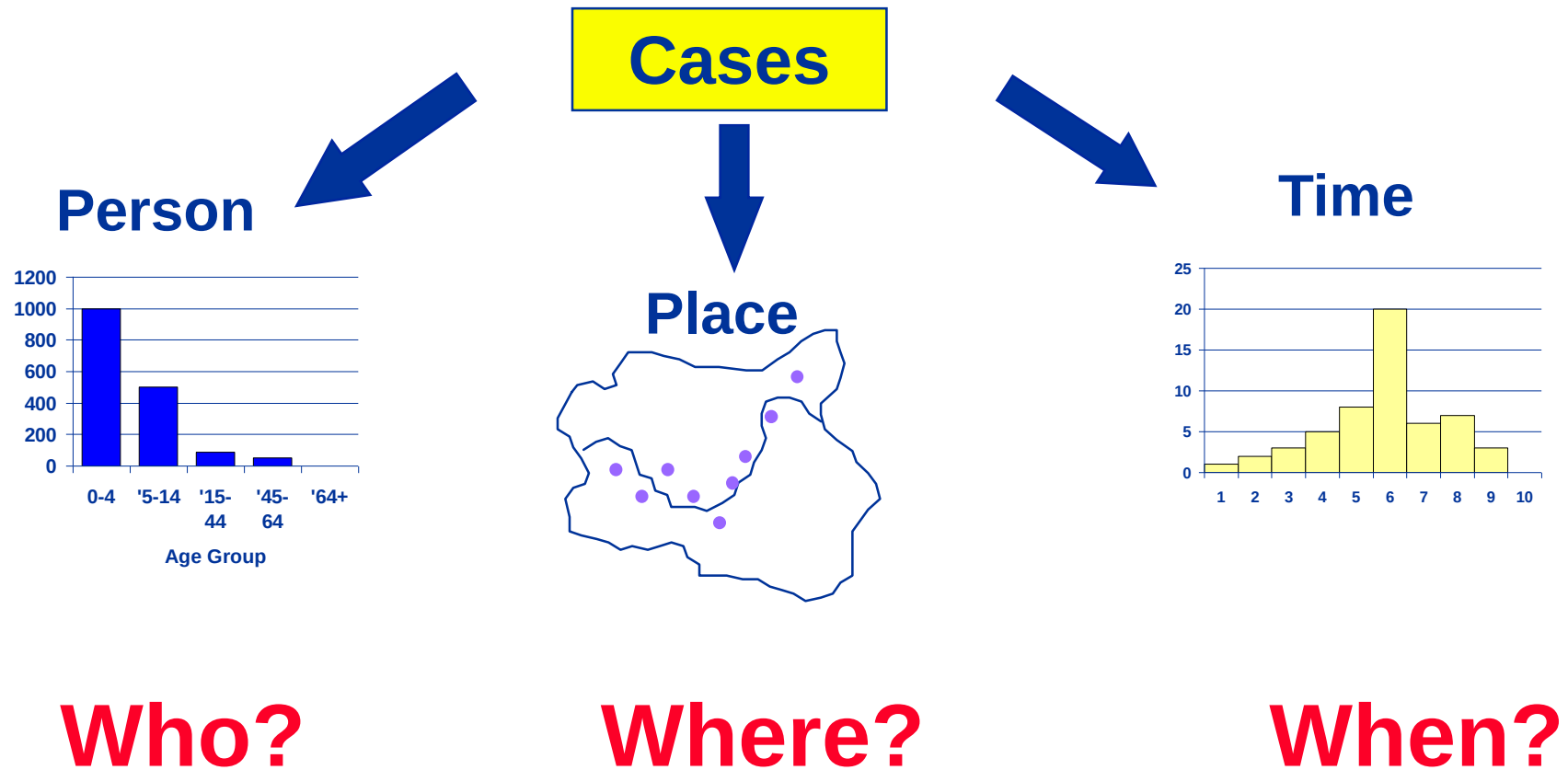
◆ **Note:**

- Describing these factors does not link them
- However can identify unusual distributions or correlations (e.g clusters)
- These insights used to generate interesting hypothesis

Hypothesis Formulation

- ◆ Person – “Who is getting the disease?”
Age, race, sex
- ◆ Place – “Where are the rates of disease highest and lowest?”
- ◆ Time – “ When does the disease occur commonly or rarely?” and “ Is the frequency of the disease now different from the corresponding frequency in the past?”

Descriptive Epidemiology



Types of Descriptive Studies

- ◆ Case report
- ◆ Case series
- ◆ Ecologic
- ◆ Cross-sectional

Case report/s

- ◆ Individual unit of observation
- ◆ Clinical case with “unusual” clinical picture
- ◆ Detailed report

Case series

- ◆ Case report leads to case series
- ◆ Case reports collected
- ◆ Group of patients with “unusual” condition
- ◆ Rare condition

Why Needed?

Case report(s), case series:

- ◆ Identification of a potentially new disease
- ◆ Alert for medication side effects
- ◆ Acts perhaps as an early warning system

Advantages

- ◆ Use available clinical data
- ◆ Detailed individual data
- ◆ Suggest need for investigation (hypothesis generation)

Disadvantages

- ◆ May reflect experience of one person/clinician
- ◆ No explicit comparison group
- ◆ Causal effect cannot be determined
- ◆ Not viewed as scientifically sound

Example

Aggressive Kaposi's sarcoma in a 6 month
old African infant: case report and
review of the literature

*Manji, K.P., Amir, H., Maduhu, I.Z. Tropical
Medicine & International Health; 5(2):85-87*

Example

Pneumocystis Pneumonia, five cases – Los Angeles

CDC. MMWR; 30(21): June 5, 1981 (original)

*Reprinted: Gottlieb, M.S. American Journal of
Public Health; 96(6):980-983, June 2006.*

Ecologic (Correlational) Studies

- ◆ Population/groups of people unit of observation
- ◆ Characteristics of the population not individual persons
- ◆ Used for between country comparisons (heterogeneity)

Ecologic Fallacy

- ◆ False assumption
- ◆ Inferences made to individuals rather than the populations/groups examined
- ◆ Common mistake

Advantages

- ◆ Simple
- ◆ Use of available group (aggregate) data
- ◆ Generate hypotheses for additional study

Disadvantages

- ◆ Unable to examine data for individuals
- ◆ Ecologic fallacy
- ◆ Persons with exposure may not be the same as the ones with disease/condition

Example

Trends in antenatal human immunodeficiency virus prevalence in Western Kenya and Eastern Uganda: evidence of differences in health policies?.

- HIV prevalence increasing Kenya
- HIV prevalence decreasing Uganda

Moore, D.M., Hogg, R.S. 2004. International Journal of Epidemiology;33:542-548.

Cross-sectional survey

- ◆ Sometimes called prevalence study
- ◆ Collection of data on several individuals
- ◆ At “one point in time”, snapshot
- ◆ Both exposure and disease examined

Advantages

- ◆ Standardized data collection tool
- ◆ Data collection focus specific location/groups
- ◆ Comparisons among study participants
- ◆ Relatively quick
- ◆ Can be repeated (trends)

Disadvantages

- ◆ Bias due to a lack of participation
- ◆ Reflects prevalence, not incidence
- ◆ Exposure and disease examined the same time

Example

Infection with HIV and intestinal parasites
among street dwellers in Gondar City,
Northwest Ethiopia April-August 2004

*Moges, F. et al. Journal of Infectious Diseases. 2006;
59:
400-403.*

Cases of Disease

The bricks from which we build
epidemiology

Case Definition

- ◆ Needed to count cases
- ◆ Used as a standard
- ◆ Aids in describing
- ◆ Can include clinical and also descriptive criteria

What is a Case Definition?

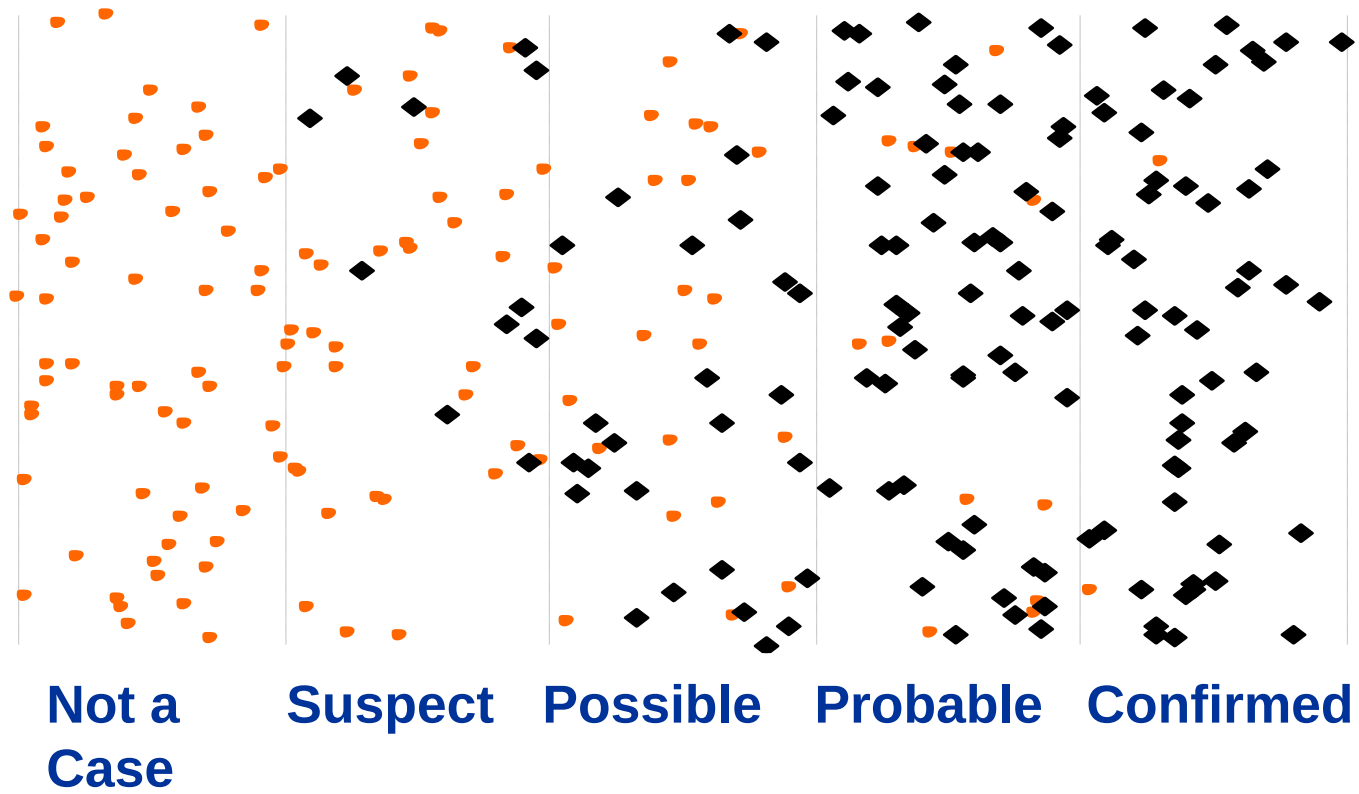
- ◆ A set of standard criteria
- ◆ Used to decide if a person has an infection or condition
- ◆ Assures that differences in health statistics are real
- ◆ Varies with the purpose

Why use a Case Definition?

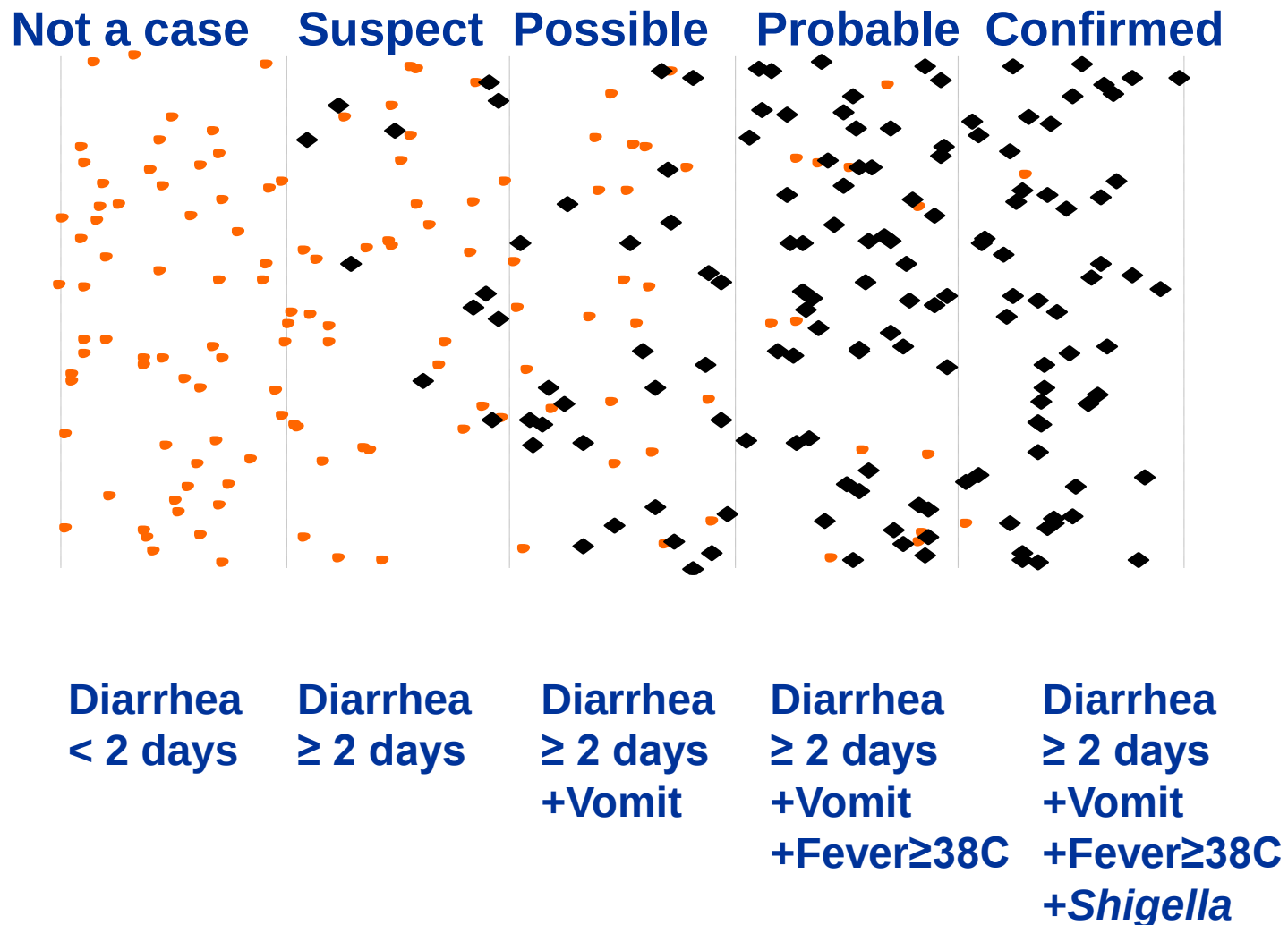
- ◆ Doctors, nurses, laboratories, and others report notifiable events
- ◆ Standardization minimizes variability in reporting.
 - aids with: monitoring trends, data quality and comparing areas

Case Definition Levels Of Certainty

- ◆ Confirmed
- ◆ Probable
- ◆ Possible
- ◆ Suspect



We Increase the Certainty That a Case is Real by Adding More Restrictive Criteria.



Examples

Surveillance CD for human rabies

◆ Clinical:

- **Encephalitis**
- **Death within 10 days of onset**

◆ Laboratory:

- **Direct FA of hair follicle, cornea or brain**
- **Anti-rabies IgG or IgM in an unvaccinated person**

◆ Epidemiological:

- **Bite from a known reservoir animal species**
- **Bite from a rabid animal**

Example: Rabies Case Definition

- ◆ Suspect
 - **Encephalitis with death in ≤ 10 days**
- ◆ Possible = suspect and
 - **animal bite**
- ◆ Probable = suspect and
 - **bite from rabid animal**
- ◆ Confirmed = suspect, possible or probable
 - **Rabies virus by direct FA from cornea, hair follicle, or brain**
 - **Or anti-rabies IgM in an unvaccinated person**

Some Data From Reported Rabies Cases

Which fit the case definitions?

Clinical	Death	Bite	Vaccine	FA	IgM
E	Y	R	N	ND	ND
E	Y	O	Y	CI -	-
E	Y	N	N	ND	+
E	N	O	Y	ND	+
E	Y	N	N	Br+	+

E=encephalitis **O= other animal**
Y = Yes **ND = Not done**
N = No **CI = Corneal Impression**
R = Rabid Animal **Br = Brain**

Measles

- ◆ Generalized skin rash of 3 to 7 days duration
- ◆ Fever $> 38^{\circ}\text{C}$
- ◆ Plus one or more of the following
 - rhinorrhea
 - cough
 - Conjunctival injection

Infectious Gastroenteritis: Case Definition

◆ Clinical:

- Diarrhea (3+ liquid stools x 2 days)
- Fever
- No other cause (drugs -- antibiotics)

◆ Confirmed Laboratory:

- Organism identified in stool

◆ Epidemiological

- Hospitalized for minimum incubation period
- In hospital during incubation

Example

WHO case definitions for AIDS surveillance in adults and adolescents

Weekly Epidemiological Record

Releve Epidemiologique Hebdomadaire

1994, 69:273-280

Example

Impact of the 1994 expanded World Health Organization AIDS case definition on AIDS surveillance in university hospitals and tuberculosis centers in Côte d'Ivoire.

Greenburg, A.E. 1997. AIDS; 11:1867-1872

Session Review

- ◆ What is the difference between analytic and descriptive study designs.
- ◆ List the four types of descriptive study designs discussed today.
- ◆ Why do we use case definitions?

What's next?

◆ Study Questions